

On Ubuntu, you will see the following code:

```
$ cd /usr/lib/ccache/
$ ll
total 24
drwxr-xr-x 2 root root 4096 May 2 13:33 ./
drwxr-xr-x 157 root root 20480 May 6 15:07 ../
lrwxrwxrwx 1 root root 16 May 2 13:33 c++ -> ../../bin/
ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 c89-gcc -> ../../
bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 c99-gcc -> ../../
bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 cc -> ../../bin/
ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 g++ -> ../../bin/
ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 g++-5 -> ../../
bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 gcc -> ../../bin/
ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 gcc-5 -> ../../
bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 x86_64-linux-
gnu-g++ -> ../../bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 x86_64-linux-
gnu-g++-5 -> ../../bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 x86_64-linux-gnu-
gcc -> ../../bin/ccache*
lrwxrwxrwx 1 root root 16 May 2 13:33 x86_64-linux-gnu-
gcc-5 -> ../../bin/ccache*
```

After installing and updating PATH variable, let's check the correctness of the compilers, as follows:

```
$ which gcc
/usr/lib/ccache/gcc
```

```
$ which g++
/usr/lib/ccache/g++
```

Let's check ccache stats:

```
$ ccache -s
cache directory          /home/vkanaujia/.ccache
primary config            /home/vkanaujia/.ccache/ccache.conf
secondary config (readonly) /etc/ccache.conf
cache hit (direct)       0
cache hit (preprocessed) 0
cache miss               0
files in cache           0
cache size               0.0 kB
max cache size           10.0 GB
```

It has stored the compiled output in the `/home/vkanaujia/.ccache` directory. We can configure ccache to store cached data in any place.

Let's check ccache's effectiveness with a sample compilation of FFmpeg code. FFmpeg is an open source library for video transcoding.

1. Download the FFmpeg source code.

```
$ wget http://ffmpeg.org/releases/ffmpeg-3.0.2.tar.bz2
```

2. Build it and time the build process.

```
$ bunzip2 ffmpeg-3.0.2.tar.bz2
$ tar xvf ffmpeg-3.0.2.tar
$ cd ffmpeg-3.0.2/
$ time make
real    12m18.544s
user    10m34.876s
sys      1m12.988s

$ ccache -s
cache directory          /home/vkanaujia/.ccache
primary config            /home/vkanaujia/.ccache/ccache.conf
secondary config (readonly) /etc/ccache.conf
cache hit (direct)       0
cache hit (preprocessed) 0
cache miss               1507
called for link          4
files in cache           4538
cache size               239.6 MB
max cache size           10.0 GB
```

3. Now, we will do a 'clean' to delete all compiled object files, as follows:

```
$ make clean
```

4. Let's rebuild sources and time the 'make', as follows:

```
$ time make
real    1m20.949s
user    1m4.824s
sys      0m5.924s
```

Here we see that with ccache, compilation is 800 per cent faster! 

References

- [1] <https://ccache.samba.org/manual.html>
- [2] <https://wiki.archlinux.org/index.php/ccache>

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